

Progress in the development of tuberculosis vaccines for cattle, goats and wildlife

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Welsh welcome



Topic: recent TB vaccine studies



- Cattle
 - BCG
 - Attenuated live vaccines
 - Subunit vaccines (DNA, protein & virus-vector)
- Goats
- Wildlife
 - Possums, badgers, wild boar and deer



Goals of vaccination against bovine TB

Reduce infection and spread of TB in cattle or wildlife species

Developing countries – prevent spread of TB in cattle

Developed countries (wildlife reservoirs of TB)

- vaccinate cattle in at risk areas (or buffer zones)
- vaccinate wildlife (maintenance hosts), if culling not possible

BCG vaccination of cattle

Advantages

- Inexpensive (low dose can be used)
- Safe
- DIVA tests to differentiate from *M. bovis* infection

Disadvantages

- Proportion of vaccinated animals react in skin test
- Protection may be incomplete
- No therapeutic effect

Summary: BCG vaccination of cattle

- Dose (10^3 to 10^6 CFU)^{1,2,5} similar protection
- Strain of BCG (Pasteur and Danish)^{8,9} similar protection
- Lyophilised v fresh culture⁹ similar protection
- Age of animal^{4,6} very young
- Pre-exposure to environmental mycobacteria^{3,7} + or -

• Ref. Buddle et al., 1995a¹;b²; 2002³; 2003⁴; 2013⁵; Hope et al 2005a⁶;b⁷; 2011⁸; Wedlock et al., 2008⁹

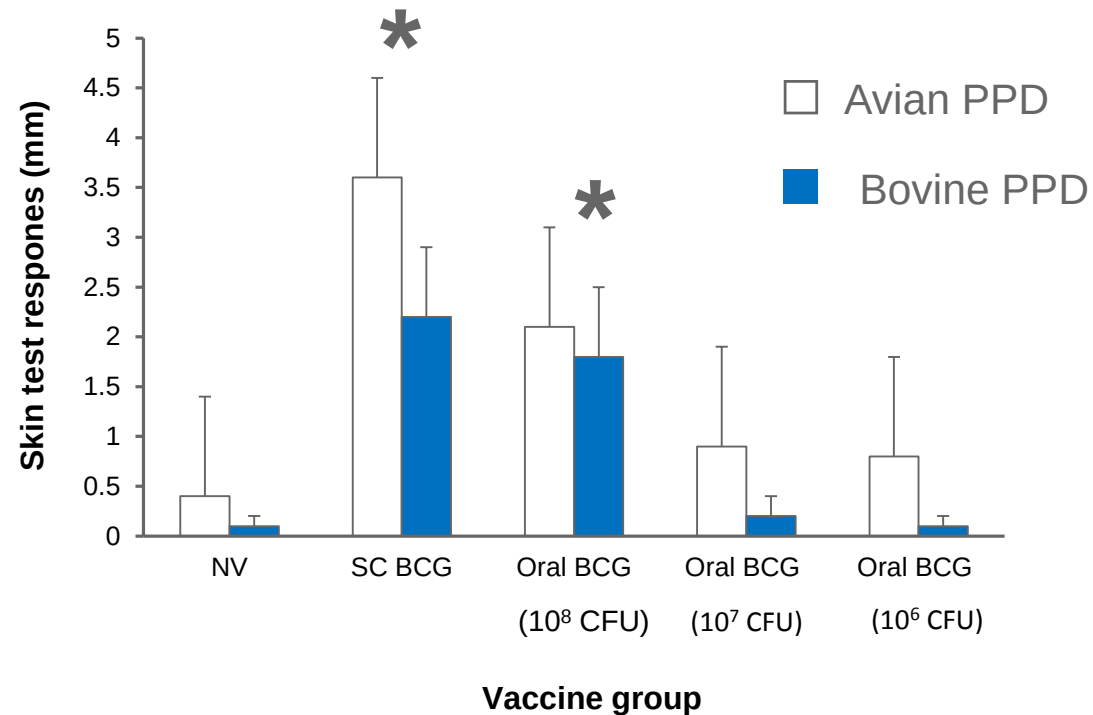
Vaccination with oral BCG

- Animals exposed to mycobacteria via mucosal surfaces
- Long history of oral immunisation with BCG in humans
- May reduce tuberculin skin test reactivity



Titration of oral dose of BCG

(Skin test; 15 weeks post-vaccination)



* Significantly protection against challenge with *M. bovis*

(Wedlock et al.,2011)

Duration of protection

12 month challenge

Median gross pathology scores (range)

Vaccine groups	Total LN score	Total lung score	Total path score
Control (n=9)	11 (0,20)	6 (0,19)	16 (0,38)
BCG 10 ⁶ CFU (s/c; n=9)	7 (0,12)	0 (0,7)*	8 (0,16)*

24 month challenge

Vaccine groups	Total LN score	Total lung score	Total path score
Control (n=9)	5 (4,10)	5 (4,10)	10 (8,17)
BCG 10 ⁶ CFU (s/c; n=9)	4 (0,9)	4 (3,9)	8 (5,17)

- Significantly reduced scores compared to that for controls, $P < 0.05$)
- Ref. Thom et al., 2012



Effect of BCG revaccination in young calves

Calf vaccine groups (n=10)

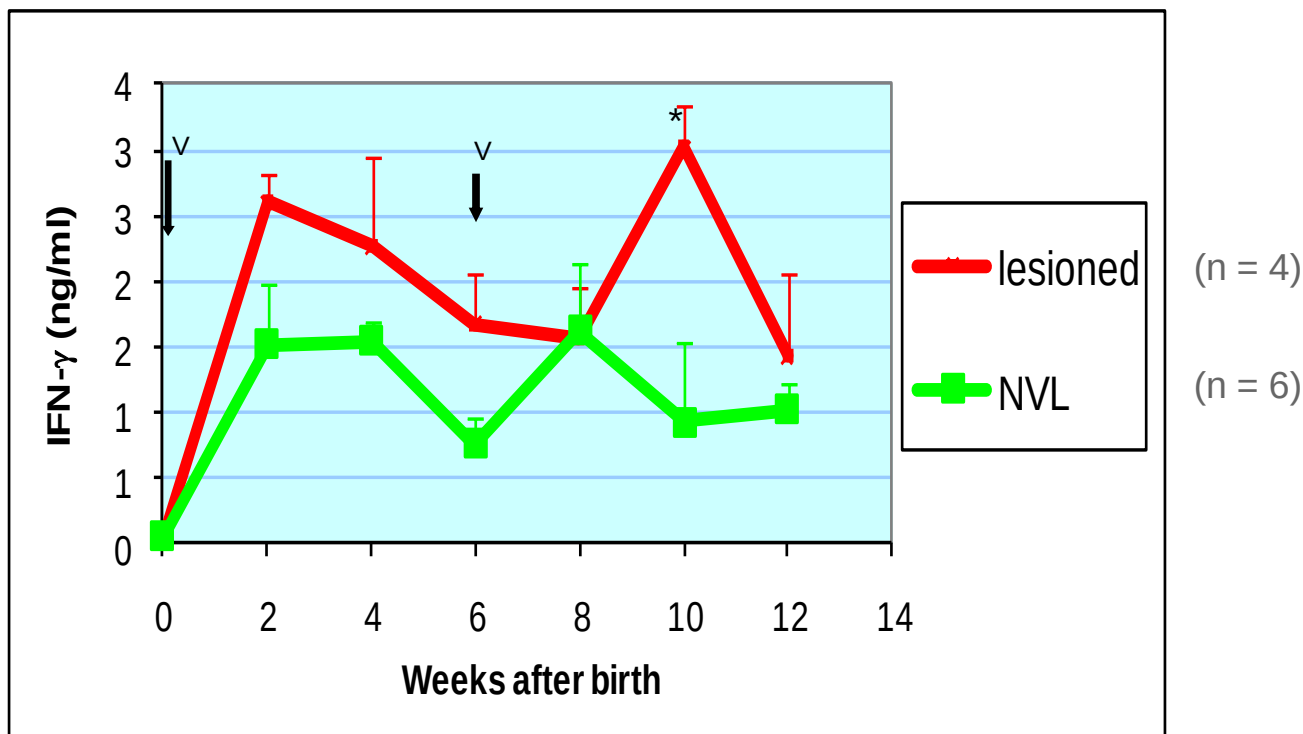
- Not vaccinated
- BCG within 8 hours of birth
- BCG at 6 weeks old
- BCG 8 hours + 6 weeks

Challenge with *M. bovis* at 14-17 weeks, necropsy 4 months later

Proportion with TB lesions

Not vaccinated	10/10
BCG within 8 hours of birth	0/10
BCG at 6 weeks old	1/9
BCG 8 hours + 6 weeks	4/10

IFN- γ responses of calves re-vaccinated with BCG



Animals which subsequently developed lesions had the highest post-vaccination IFN- γ responses

Long term effects of BCG vaccination and can immunity be boosted



Vaccine groups (total 79 calves)

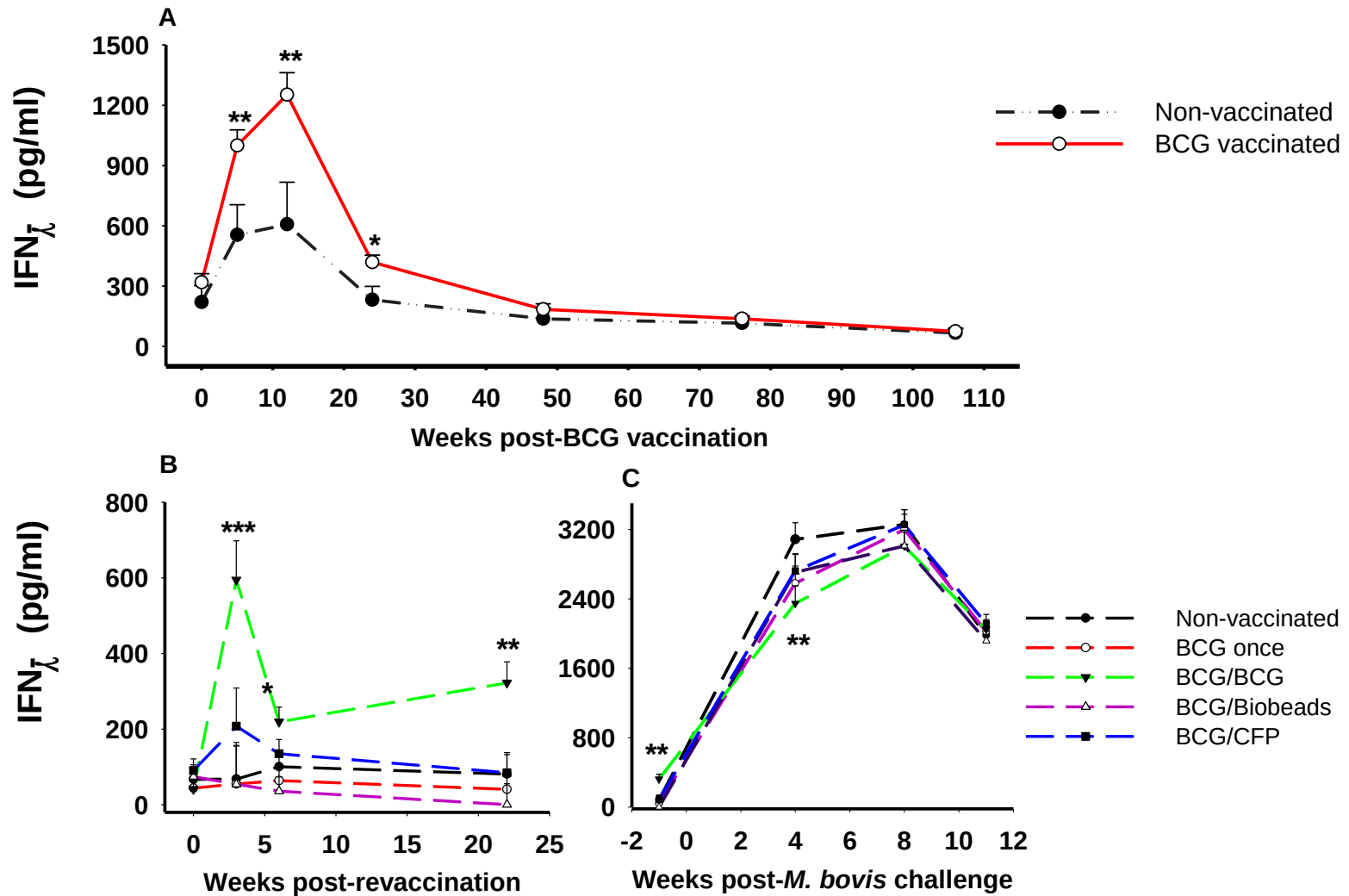
1. Non-vaccinated (n=17)
2. S/c BCG (n=16)
3. S/c BCG, at 2 years revaccinate BCG (n=15)
4. S/c BCG, at 2 years revaccinate with *M. bovis* culture filtrate protein (CFP)/adjuvant (n=15)
5. S/c BCG, at 2 years revaccinate with biobeads displaying mycobacterial proteins, ESAT-6 and Ag85A on the surface (n=16)

Vaccinate calves at 2-4 weeks of age

Revaccinate some groups at 2 years of age

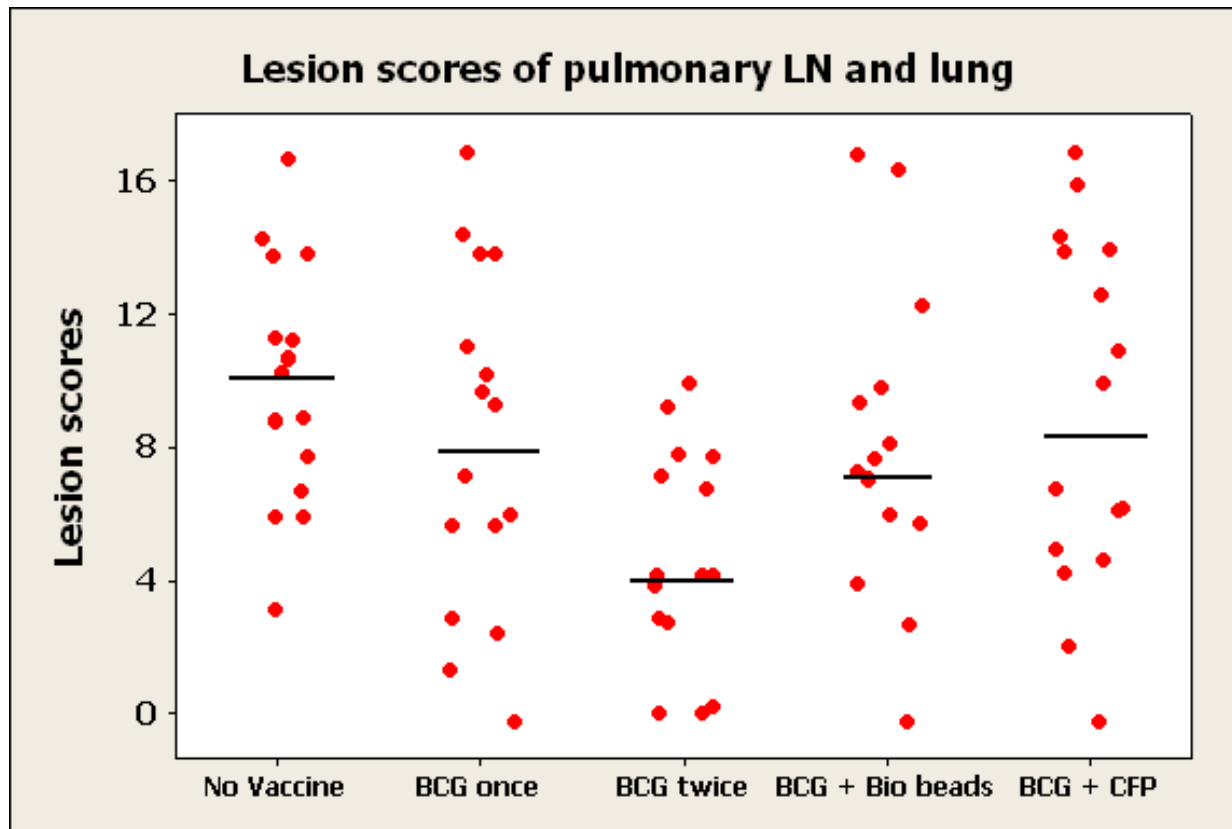
Challenged with TB at 2½ years of age and slaughter 3 months later

Whole blood IFN- γ responses to bovine PPD



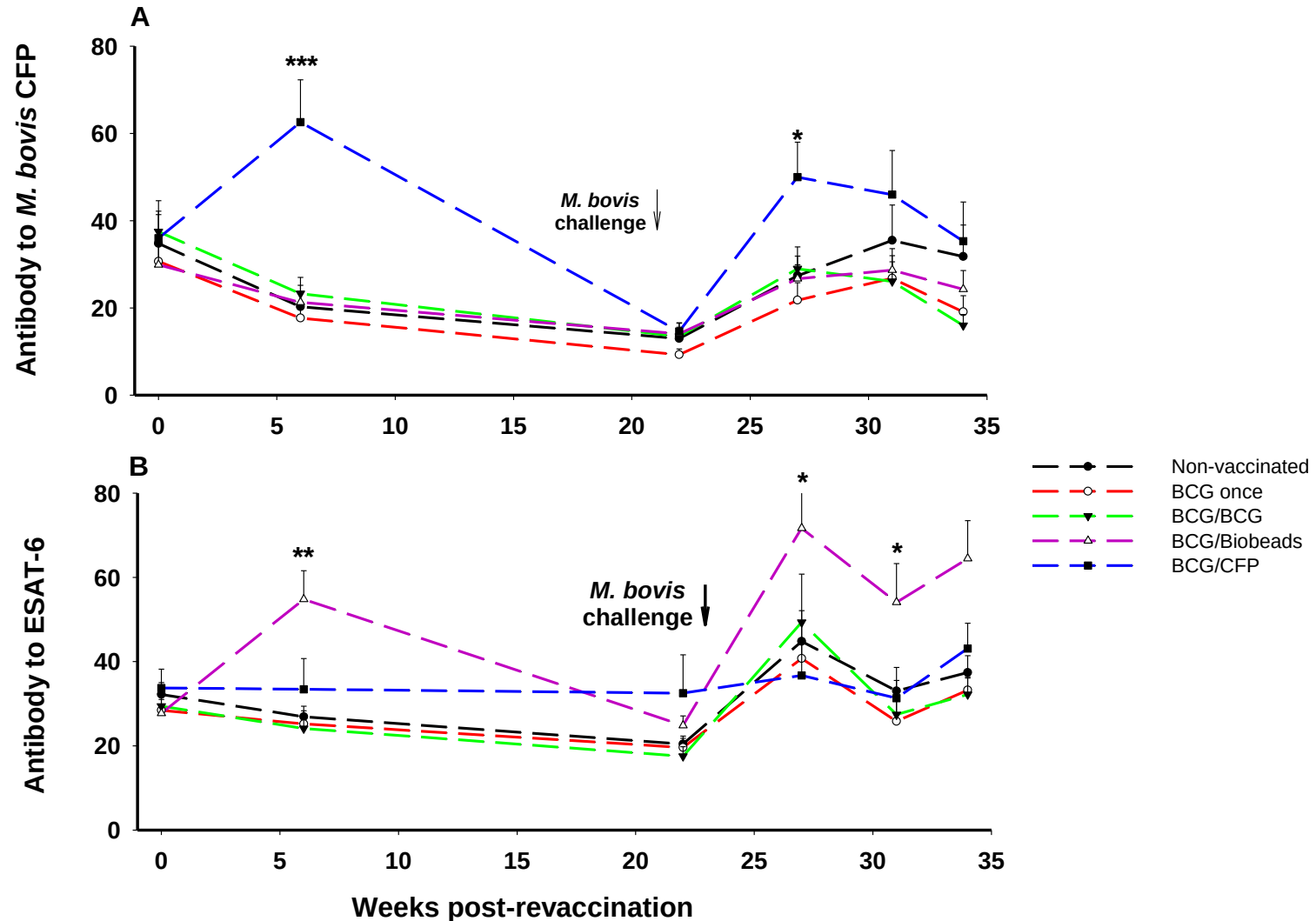
Significant different to the non-vaccinated group, * P < 0.05, ** P < 0.01, *** P < 0.001

Total lung and pulmonary lymph node lesion scores following challenge with *M. bovis*



BCG-revaccinated group had significantly lower lesion scores than no vaccine group ($P < 0.001$)

Serum antibody responses to *M. bovis* CFP and ESAT-6



Significant different to the non-vaccinated group, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Field BCG trial in cattle - Mexico



■ Vaccinated 70 one to two week old calves, BCG s/c
Equivalent number non-vaccinated

■ Followed until 12 months of age

■ Positive case of TB defined by the following tests

- Tuberculin skin test
- IFN- γ PPD-B
- IFN- γ ESAT-6/CFP10

BCG-vaccinated group:

6 of 65 classified as TB-infected

Non-vaccinated group:

15 of 66 classified as TB-infected

Field BCG trial in cattle - Ethiopia

- Calves < 3 wks old vaccinated 10^6 CFU BCG s/c
- In contact with reactor cattle from 3 mths post-vaccination
- In-contact period 10-22 mths, then killed and necropsied

	Gross pathology (VL) % (n)	<i>M. bovis</i> culture +ve % (n)	Spread outside head and lung regions % (n)	Condemned at meat inspection % (n)
Naïve (n=14)	86 % (12)	79 % (11)	21 % (3)	71 % (10)
BCG (n=13)	38 % (5)	31 % (4)	0 % (0)	23 % (3)
P-values (Fisher's exact test)	0.018	0.021	NS	0.021

Cattle vaccination trial:

Muzzle Station (LandcareResearch, NZ)

- Isolated farm
- TB incidence 5 -10% of cattle/yr

BCG vaccination trial

- Five cohorts of 'free ranging' cattle, skin-tested, +ves excluded.
- Approx. half vaccinated with BCG orally (mostly 10^8 CFU)
- Cattle inspected for TB at slaughter 1-3 yr later.



Progress results

Provisional diagnoses, some cultures pending

Cohort birth year	Oral BCG Dose	Vaccinates	Non Vaccinates	<i>P</i> 2 x2 contingency table
2.5 yr cattle 2006	10 ⁸	0/30 (0.00%)	5/130 (3.85%)	0.58
1.5 yr cattle 2007	10 ⁸	5/172 (2.91%)	8/118 (6.78%)	0.15
Weaners 2008	10 ⁸	11/177 (6.21%)	12/85 (14.12%)	0.039*
Weaners 2009	10 ⁸	10/167 (5.88%)	21/106 (19.81%)	<0.001***
Weaners 2010	2 x 10 ⁷	2/98 (2.04%)	7/84 (8.33%)	0.083
Total		28/644 (4.35%)	53/523 (10.13%)	<0.001***

Differentiate BCG-vaccinated from *M. bovis*-infected cattle (DIVA tests)

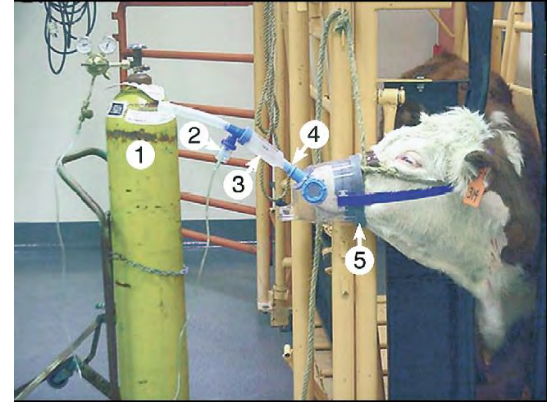


- Whole blood IFN- γ test (ESAT-6, CFP10, Rv3615c)¹⁺
- Differential skin test (ESAT-6, CFP10, Rv3615c, Rv3020c)
 - Recombinant proteins or peptides^{2,3}
 - Proteins displayed on biobeads⁴

Ref. Vordermeier et al., 2011¹; many other references⁺; Whelan et al., 2010²; Jones et al., 2012³; Chen et al., 2014⁴

Attenuated mycobacterial TB vaccines

Attenuated *M. bovis* vaccines



- Attenuated *M. bovis* $\Delta RD1$ mutant¹
 - Vaccination reduced TB pathology and bacterial counts (thoracic LNs)
 - Protection similar to that from vaccination with BCG
 - Neither vaccine induced IFN- γ ESAT-6/CFP10 prior to challenge
- Attenuated *M. bovis* $\Delta mce2$ double deletion mutant²
 - Lower histopathological lesion score for lungs and LNs v BCG group ($P < 0.05$)
- BCG strain overexpressing Ag85B³
 - Lower histopathological lesion score for lungs v BCG group ($P < 0.05$)
- BCG $\Delta zmp1$ mutant (Zmp1 inhibit phagosome-lysosome fusion)⁴
 - Superior T cell memory responses v BCG

TB DNA vaccines

DNA vaccines for cattle

- DNA alone – little protection
- DNA vaccines + immunopotentiators → some protection
 - DNA + co-stimulatory molecules (CD80 and CD86) (Maue et al., 2004)
 - DNA + adjuvant (DDA) (Cai et al., 2005)
- DNA prime/ BCG boost better than BCG alone
(Skinner et al., 2003; Cai et al., 2006)
- Both DNA/BCG or BCG/DNA effective
(Skinner et al., 2005)

TB protein vaccines

TB protein vaccines



- Proved difficult to induce protection against TB in cattle with protein vaccines
- Co-administration of BCG and *M. bovis* culture filtrate vaccine at adjacent sites s/c induced improved protection v BCG alone^{1,2}
- Display of antigens on particles (Biobeads)
- Use of Toll-like receptors to induce a CMI response
 - TLR2 – Pam3Cys, Pam3CSK4, mycobacterial PIMs²
 - TLR4 – Lipid A, glucopyranosyl lipid A³
 - TLR7/8 – Resiquimod³
 - TLR9 – CpG oligonucleotides¹

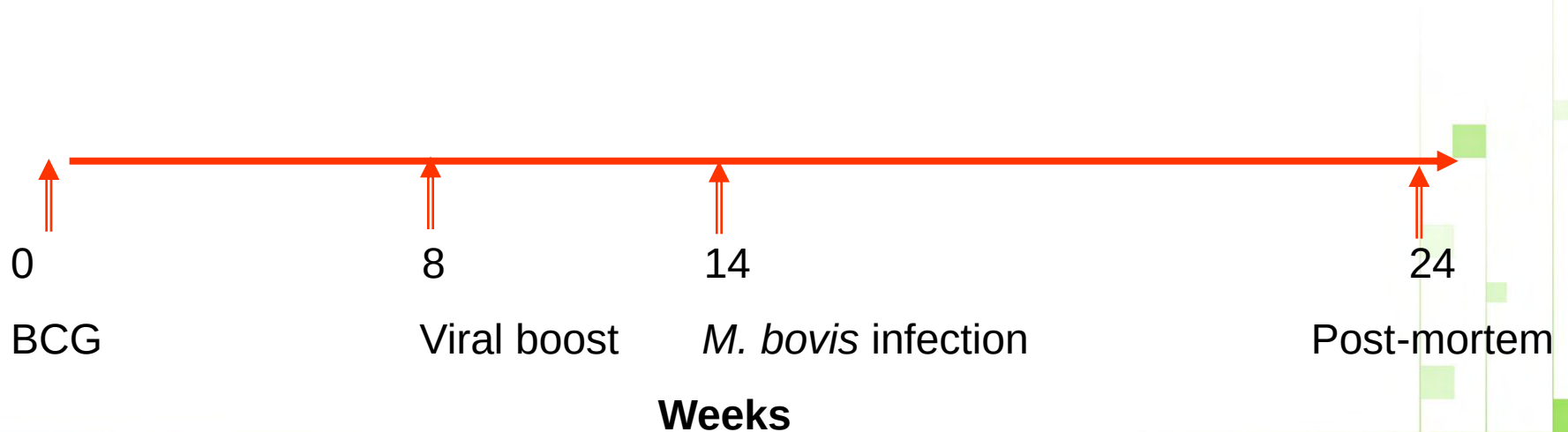
Ref. Wedlock et al., 2005¹; 2008²; Jones et al., 2014³

Virus-vectored TB vaccines

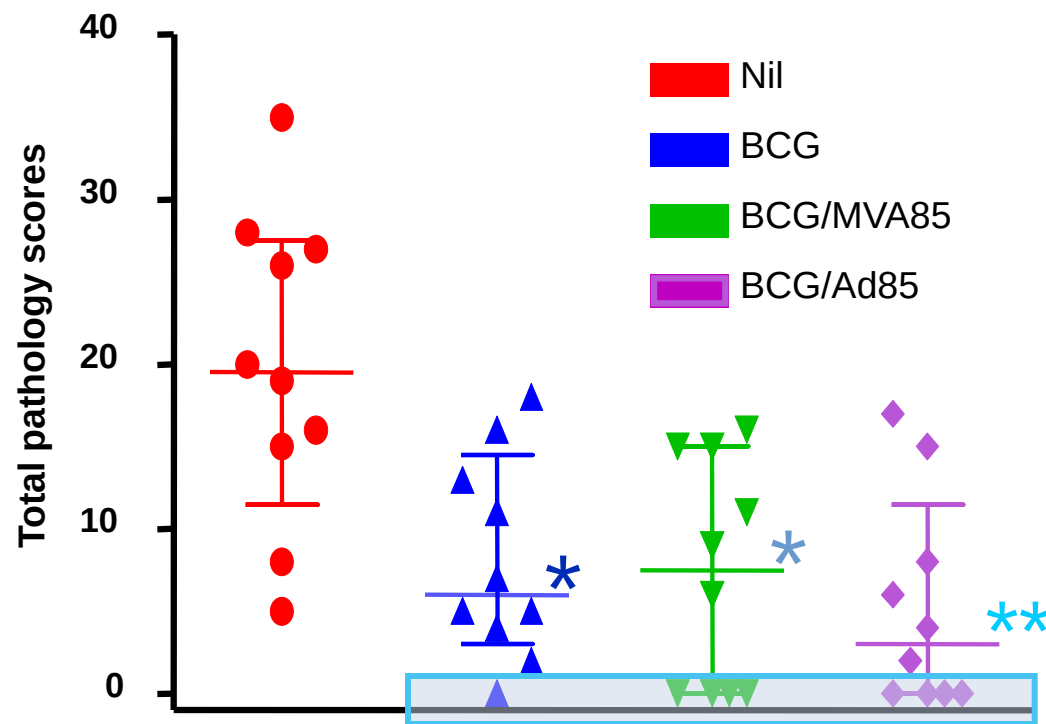
Testing Ad85A and MVA85A in a cattle challenge experiment (Vordermeier et al., 2009)

Vaccination/challenge doses/routes:

- MVA85A: 1×10^9 pfu, i.d.
- Ad85A: 2×10^9 pfu, i.d.
- BCG (SSI): 10^6 CFU, s.c.
- *M. bovis* (AF2122): 2000 cfu, intratracheal route



Viral vector vaccines reduced pathology with no visible (or histological) signs of infection in a proportion of animals



Further studies with virus-vector vaccines

- Boosting with adenovirus 5 (Ad5) expressing Ag85A provided better protection than Ad5 expressing 4 mycobacterial proteins (Ad5-TBF)¹
- Dose and route of immunisation with Ad5-TBF²
 - Strongest IFN- γ responses
 - 2×10^9 infectious units
 - Delivered intradermally

Ref. Dean et al., 2014a¹; b²

Summary of TB vaccines for cattle

- New TB vaccines for cattle are promising, but no single vaccine is better than BCG
- Combinations of BCG + other TB vaccines (virus vector, protein or DNA) induced better protection than BCG alone
- BCG vaccine shown to reduce disease in experimental and field trials
- Revaccination with BCG – effective when immunity has waned
- Effective non-sensitising TB vaccine for cattle?

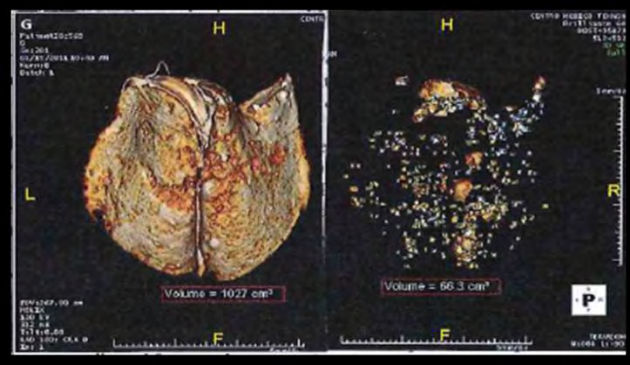


Caprine tuberculosis



- Goats are susceptible to *M. caprae* and *M. bovis*
 - no difference in the pathology
- *M. caprae* infection of goats
 - *M. caprae* may cause infections in cattle and humans, and problem for bovine TB diagnosis
 - Cavities in granulomas (similar to pathology in humans)
- Development of a vaccine against caprine TB
 - Assist in control of this disease in goats
 - Valuable model for development of TB vaccines in cattle and humans

Vaccination of goats against TB



- Experimental challenge - low dose of *M. caprae* endobronchially¹
- BCG vaccination (s/c) significantly reduced pathology and bacterial loads^{2,4}
- BCG prime/ adenovirus (Ad) - Ag85A or Ad - Ag85A/TB10.4/TB9.8/Acr2 boost improved protection ^{2,4}
- DIVA reagents not compromised by vaccination of goats with BCG, BCG/Ad or Johne's disease (JD) vaccine
- Vaccination of goats with JD vaccine → partial protection against *M. caprae*³
- Aerosol challenge of goats with *M. bovis* → lung and LN pathology⁵

TB vaccines for wildlife



Brushtail possums - New Zealand



White-tailed deer - USA



Badgers - UK and Ireland



Wild boar - Spain



Oral BCG vaccine for possums

- Encapsulation in a lipid matrix,
 - Small numbers of BCG shed in faeces¹
 - Protect against experimental aerosol challenge with *M. bovis* decrease severity of disease²
 - Protection wanes between 6 to 12 months³
- Baits stable for 3-5 wks field conditions⁴
- 85 and 100% of possums accessed baits at bait densities 40-80 sachets/ha⁴

Ref. Wedlock et al., 2005¹; Aldwell et al., 2003²; Buddle et al., 2006³; Cross et al., 2009⁴



Protection against natural exposure to *M. bovis* in possums



- Approx. 50% of possums orally vaccinated with BCG

- Trap every 2 mths and check for TB lesions

- Kill out all possums after 2 yrs and necropsy

- Proportion with TB lesions

12/71 control v. 1/51 vaccinated ($P < 0.05$)



Ref. Tompkins et al., 2009



Duration of protection

- Duration of protection following a single oral BCG¹
 - Possums from a TB-free area were orally vaccinated with BCG released back into the field
 - 28 months later, vaccinated and non-vaccinated possums captured, relocated to a PC3 unit and challenged s/c in the paw
 - BCG-vaccinated group, significant reduction in LN bacterial counts

Ref. Tompkins et al., 2013



Vaccination of badgers with BCG



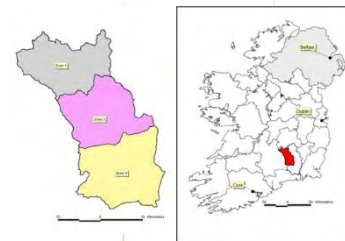
- BCG vaccine shown to be safe in badgers¹
- Vaccination with BCG via s/c or mucous membranes³, I/mus² or oral BCG in lipid matrix⁴ protected against *M. bovis* challenge
- Comparison of oral vaccination with Pasteur and Danish BCG strains induced similar levels of protection against *M. bovis* challenge⁵
- Ready uptake of buried baits, 51% of captured badgers after 7 days⁶
- Modelling – revealed that combination of vaccination and culling may be more effective than either strategy alone⁷

- Ref. Lesellier et al., 2006¹; 2011²; Corner et al., 2008³; 2010⁴; Murphy et al., 2014⁵
Palphramand et al., 2012⁶ ; Smith et al., 2012⁷

Field BCG badger trials



- Field trial, 3 years duration (2006-09)¹
 - 179 vaccinated with 10^6 CFU BCG I/mus, 83 non-vaccinated
 - Vaccination reduced incidence of seroconversion by 74%
- Follow up from trial, risk of unvaccinated cubs testing positive to a series of TB tests reduced significantly as proportion of vaccinated individuals in the social group increased²
- Irish 3-year field trial –
 - 3 zones: 100%, 50% & 0% vaccination with oral BCG
 - Annual vaccination: Vaccine/placebo blind coded
 - Capture/release, monitor serology, final necropsy all
- On-going trials in Wales and England (licenced I/mus BCG vaccine)
 - Ref. Chambers et al., 2011¹; Carter et al., 2012²



Summary: badger vaccination



- Convincing evidence implicating badgers in spread of TB to cattle
- Culling and segregation in long term may be unsustainable for ethical, economic and practical reasons
- Vaccination is a practical method provided it is efficacious and cost-effective
- I/mus vaccination – useful to determine efficacy, practicality, costs and building confidence in the principle of vaccination
 - Issues – high cost, selection of animals trapped, welfare issues
- Oral bait – cost effective method for vaccine delivery, wide coverage
 - Issues – identify delivery system, proportion vaccinated not known, non-target species



BCG vaccination of white-tailed deer

- Protection of white-tailed deer against intratonsillar challenge with *M. bovis*
 - Orally with 10^9 CFU of BCG in lipid matrix or PBS¹
 - Orally with 10^8 CFU of BCG in PBS²
 - Subcutaneously with 10^6 CFU Pasteur or Danish strains of BCG³
- Subcutaneous revaccination with BCG after 6 weeks did not enhance protection⁴, in contrast to studies in red deer⁵

Ref. Nol et al., 2008¹; Palmer et al., 2007²; 2009³, 2014⁴; Griffin et al. 1999⁵



Persistence and transmission of BCG in deer

■ Persistence of BCG in vaccinated deer

- BCG recovered from lymphatic tissue of deer at 1, 3, 6, 9, 12 months after oral vaccination with 10^9 CFU of BCG, but not from muscle nor from any tissues after a 10^8 CFU oral dose²
- BCG recovered from lymphatic tissue of deer up to 9 months after s/c vaccination with 10^6 CFU, but not from muscle¹

■ Transmission of BCG²

- Immunological evidence of transmission of BCG to non-vaccinated deer co-housed with deer orally vaccinated with 10^9 CFU of BCG, but not to cattle which alternatively shared pen space with deer

Ref. Palmer et al., 2010¹; 2012²; Nol et al., 2013³



TB in Eurasian wild boar

- Wild boar – wildlife reservoirs for *M. bovis*, high densities in hunting estates and national parks (often >40 % TB prevalence in dry Mediterranean sites)
- TB in wild boar identified in at least 10 European countries
- Lesions predominantly found in head LNs (mandibular), may become generalised with lung lesions (in 50% of cases)
- Role of wild pigs as a reservoir for *M. bovis* ?





Vaccination of wild boar



- Selective feeders for wild boar piglets (2-4 months of age) and apply baits in early and late summer in south-central Spain¹
- Vaccine baits- cereal-based matrix containing a capsule to deliver vaccine²
- Survival of BCG in baits³
 - No loss of viability up to 36 h in the field (temperature range 11 to 41C°)
 - Loss of 2 logs by 24 h with storage at 37°C in laboratory
- After oral BCG vaccination of 10^5 CFU, no isolation from tissues of piglets or from faeces from 1-71 days after vaccination³

Ref. Ballesteros et al., 2011¹; Beltrán-Beck et al., 2012²; 2014³)

Evaluation of TB vaccines in wild boar

- Challenge model established – 10^4 to 10^6 CFU via oropharyngeal route¹
- Vaccination - oral live BCG (10^5 CFU), oral or parenteral heat-inactivated *M. bovis* vaccine IV (10^6 bacilli)²
 - Lower pathology and bacterial loads, but not significant different v controls
- Revaccination – oral live BCG (10^6 CFU, 52 day interval)³
 - Significantly lower pathology and bacterial loads v controls
- Revaccination – oral IV (10^6 CFU, 52 day interval)⁴
 - Significantly lower pathology and bacterial loads v controls
- Ongoing field trial of oral BCG and IV
 - Started in 2012. 1st two years yield significant reduction of piglet lesion scores in IV treatment sites v controls

Ref. Ballesteros et al., 2009, Garrido et al., 2011; Gortazar et al., 2014; Beltrán-Beck et al. 2014b



Future directions for wildlife TB vaccines

- Improvements in oral bait formulation
- Systems to avoid uptake by non-target species (attractants and delivery systems)
- New improved TB vaccines (killed or attenuated *M. bovis*)
- Therapeutic TB vaccine for wildlife?
- Field trials to demonstrate prevention or reduction of spread of *M. bovis* infection to domestic animals

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Questions?

